

Consider a standard deck of 52 cards, as displayed in the following figure. In the first and third rows all cards are black; in the second and fourth rows, all cards are red.



Consider the following events when a card is randomly selected.

- $A$ : card selected is a heart.
- $B$ : card selected is the color black.
- $C$ : card selected is a face card (i.e. J, Q, K)
- $D$ : card selected is a King.

1. Find  $P(A)$ .

2. Find  $P(C)$ .

3. Find  $P(D^C)$ .

4. Find  $P(A \cup B)$ .

Consider the seven sided die (with sides 0, 1, 2, 3, 4, 5 and 6) from your reading questions. You roll this die once. Let:

- $A$  be the event that a given roll yields an even number.
- $B$  be the event that a given roll is greater than or equal to three.
- $C$  be the event that the number appears in the phrase "Stat 20".

5. Find  $P(A)$ ,  $P(B)$  and  $P(C)$

6. Find  $P(B \cap C)$

7. Find  $P(A \cap B^C)$ .

One number will be drawn at random from each of the two sets below:

$$A = \{1, 2, 3\}; B = \{1, 2, 3, 4\}$$

8. What is the probability the number drawn from  $A$  is greater than the one drawn from  $B$ ?

9. What is the probability that the number drawn from  $A$  is equal to the one drawn from  $B$ ?

Consider an outcome space  $\Omega$  with events  $A, B$  that are *not* mutually exclusive. Let  $P(A) = 0.5$ ,  $P(B) = 0.7$  and  $P(A \cap B) = .4$ .

10. Draw a Venn Diagram containing two circles, one representing  $A$  and the other  $B$ .

11. Shade in the space in the diagram above corresponding to event  $A$ .

12. Shade in the space in the diagram above corresponding to event  $B$  in a way that is different than the way you shaded in the space for event  $A$ .

13. Based on your shading, calculate the probability  $P(A \cup B)$ . Explain why you cannot add  $P(A)$  and  $P(B)$  directly to obtain this answer.